

09

A person with diabetes is in diabetes remission if their blood glucose concentration is below the diabetes threshold concentration for at least 3 months. This diabetes remission is achieved without taking medication.

Scientists investigated whether a weight-loss programme would result in type II diabetes remission.

The scientists:

- used a computer-generated list to select 380 volunteers from a large number of health centres
- selected volunteers aged 25 to 60 years, each with less than 5 years duration of type II diabetes
- divided the volunteers in the ratio 1 : 1 between experimental group **P** and control group **Q**
- placed group **P** on a weight-loss programme for the 2-year duration of this investigation
- recorded loss of mass and percentage of volunteers in each group in type II diabetes remission after 2 years.

09.1

Give **two** reasons why a weight-loss programme could be used to treat type II diabetes but **not** type I diabetes.

[2 marks]

1 _____

2 _____

Question 9 continues on the next page

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The design of this investigation helps to support the validity of any conclusions obtained.

Suggest and explain **three** features of this investigation that justify this statement.

[3 marks]

1 _____

2 _____

3 _____

0 9 . 3 Table 3 shows some of the scientists' results.

Table 3

	Group P	Group Q
Percentage achieving diabetes remission	42.1	4.7
Percentage achieving weight loss ≥ 15 kg	15.8	2.1
Percentage with weight loss ≥ 15 kg achieving diabetes remission	70.0	75.0
Percentage with weight gain achieving diabetes remission	1.9	1.9

Use the information provided to calculate the difference in number of volunteers between the two groups who achieved diabetes remission.

Show your working.

[2 marks]

Answer _____

Question 9 continues on the next page

Use all the information to evaluate whether a weight-loss programme is an effective treatment for type II diabetes.

[4 marks]

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